

Predicting the long-term impact of AI assistants on collective intelligence

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Abstract

Recent advances in artificial intelligence (AI) have enabled widespread use of personal AI assistants for various cognitive tasks, ranging from simple everyday tasks to scientific discovery and creative writing. While AI assistance can improve the performance of each individual, it can weaken their subsequent performance without AI and further limit the diversity of interests and opinions. These risks raise the question of whether human collective intelligence can persist as our society becomes increasingly reliant on AI assistants. To address this question, we develop an evolutionary model of collective intelligence, where each individual participate in a prediction task, consult AI, and imitate each other's prediction strategy. We find that "Chatbot" AI, which depends on users' interests, can prevent the emergence of collective intelligence by either limiting diversity or propagating systematic bias through overreliance. Directly incentivizing or penalizing AI use generally worsens these outcomes. Instead, a balanced incentive structure that preserve minority expertise and reward contributions to collective outcome can promote the emergence of collective intelligence. However, the inherent ability of the population's to make prediction in the absence of AI remains suboptimal even under this balanced incentive structure, raising concerns for continued AI reliance. **[SWP: conclusion?]**

27 Introduction

28 Collective intelligence (CI) describes a phenomenon in which a group’s ability to
29 solve problems, make decisions, or adapt to changing environments exceeds each
30 member’s ability (Galton, 1907; Woolley et al., 2010; Yaniv and Milyavsky, 2007;
31 Hommes et al., 2005; Lorge et al., 1958). This phenomenon, also referred to as
32 wisdom of crowds, has thus served as a central mechanism underlying the success of
33 diverse organizations, ranging from animal groups (Dreyer et al., 2025; Simons, 2004;
34 Berdahl et al., 2013) and human societies (Woolley et al., 2010; Riedl et al., 2021) to
35 machine-learning systems (Breiman, 1996, 2001). Understanding the conditions that
36 promote and maintain CI therefore remains a fundamental question across social
37 sciences (Mann and Helbing, 2017), evolutionary biology (Wang et al., 2025), and
38 computer science (Werfel, 2026). Addressing this question has become particularly
39 timely as artificial intelligence (AI) has begun to reshape how individuals acquire
40 information, make decisions, and contribute to collective outcomes (Cui and Yasserli,
41 2024).

42 The recent emergence of generative AIs and large language models (LLMs) has
43 changed the way individuals engage in various cognitive tasks, ranging from simple
44 everyday tasks to more complex tasks (Noy and Zhang, 2023; Brynjolfsson et al.,
45 2025), including scientific discovery (Lu et al., 2026) and creative writing (Doshi
46 and Hauser, 2024). In particular, while the use of generative AI models, such as
47 ChatGPT, has been rapidly increasing (Bick et al., 2026; Sajadieh et al., 2026), the
48 potential long-term impact of personal AI assistants on human society, especially the
49 CI, remains uncertain. Some argue that AI can promote CI by enhancing individual
50 expertise (Vaccaro et al., 2024), and opinion aggregation (Burton et al., 2024; Tessler
51 et al., 2024). Conversely, empirical evidence indicate that the use of AIs can limit the
52 diversity of opinions (Fügener et al., 2021; Doshi and Hauser, 2024; Lee and Chung,
53 2024; Meincke et al., 2025), a key requirement for the emergence of CI (Hong and
54 Page, 2004; Lorenz et al., 2011; Aplin et al., 2014; Becker et al., 2017).

55 Our ability to predict the long-term impact of personal AI assistants on human
56 CI is further hampered by the apparent discrepancy in the incentives that shape AI
57 uses. On one hand, many industrial sectors are pushing towards increased AI usages,
58 where the number of “tokens” used is often perceived as a proxy for productivity
59 (Bousquette, 2026; Keary, 2026). On the other hand, many academic institutions
60 appear to discourage, and even penalize, AI uses, reflecting concerns about plagia-
61 rism, misinformation, and limited independence (Thorp, 2023; Chawla, 2026). Such
62 discrepancy raises a question about whether directly incentivizing or penalizing AI
63 usage will benefit or harm human CI. These observations further highlight the im-
64 portance of indentifying a robust incentive structure that can promote CI in the
65 presence of personal AI assistants. Such incentive structure should seek to retain
66 the benefits of AIs and prevent potential risk, such as preserving the diversity of
67 individual opinions.

68 To address these questions, we extend previous approaches for modeling CI, where

each individual participates in a prediction task from partial observations of underlying causal factors (Mann and Helbing, 2017; Wang et al., 2025; Hong et al., 2012). In particular, we allow each individual to independently consult with personal AI assistants and revise their predictions accordingly. The resulting model allows us to investigate the robustness of human CIs under different assumptions about the AI model’s ability to make predictions as well as incentive structures that are associated with the prediction task. In doing so, we assess the impact of directly incentivizing or penalizing AI uses on CI and further suggest a balanced incentive structure that can maintain CI in the presence of AI.

Mathematical model

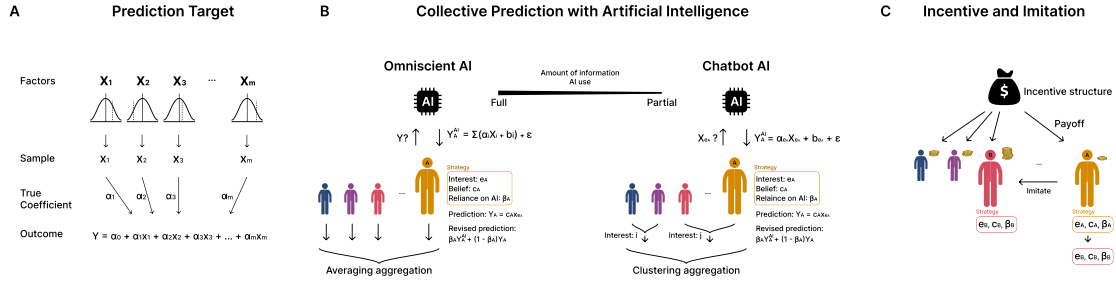


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of personal AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

We begin by extending previously established models of CI, in which a group of players participates in a prediction task (Figure 1): each player makes an individual prediction based on partial observations of the underlying causal factors, and the group forms a collective prediction by aggregating these individual predictions (Hong et al., 2012; Mann and Helbing, 2017; Wang et al., 2025). Specifically, the prediction target, Y , is modeled as a linear combination of M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor is drawn from independent normal distribution (Figure 1A). Then, each individual A observes a single factor of their interest x_{e_A} and makes a prediction based on their own belief, c_A : $Y_A = c_A x_{e_A}$ (Figure 1B). We assume each individual can then independently consult their AI assistants and revise their prediction based on their reliance on AI, β_A :

90 $Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A$ (Figure 1B). The revised predictions are then aggregated
 91 using two alternative aggregation rules, averaging and clustering, to form collective
 92 prediction.

93 To model AI, we assume that AI has already been trained on similar tasks and is
 94 able to make predictions based on partial or full observations of causal factors (Figure
 95 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on
 96 its internal parameters, which can include factor-specific bias (b_i) and stochastic
 97 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 98 amount of information AI uses when making predictions. In one extreme, the AI
 99 can observe all causal factors and use all information to make predictions, regardless
 100 of the user’s interest: $Y_A^{\text{AI}} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 101 like a chatbot, where the AI predicts the outcome based only on the user’s interest:
 102 $Y_A^{\text{AI}} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’
 103 and ‘Chatbot AI’, respectively.

104 We assume that appropriate aggregation rule for collective prediction depends
 105 on the type of AI. When Chatbot AI is used, each revised prediction depends on a
 106 single factor. Thus, we use clustering aggregation, under which revised predictions
 107 are first averaged among individuals who are interested in the same factor, and
 108 then summed across factors. When Omniscient AI is used, however, each revised
 109 prediction contains information about all causal factors, in which case clustering
 110 these predictions by interest would be inappropriate. Instead, we use averaging
 111 aggregation, under which revised individual predictions are averaged directly to form
 112 the collective prediction.

113 Each generation consists of multiple round of predictions, and each player is re-
 114 warding a payoff according to the incentive structure (Figure 1C). For simplicity,
 115 we assume that a sufficiently large number of prediction rounds occurs within each
 116 generation, allowing us to use the expected payoff value rather than computing the
 117 payoffs for every round. We consider two incentive structures from previous work,
 118 “feedback” and “niche-expert”, which have been shown to promote near-perfect col-
 119 lective accuracy in the absence of AI (Wang et al., 2025). The feedback structure
 120 gives higher payoff to individuals who can push the collective prediction toward the
 121 true target, regardless of their individual predictive accuracy: $\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})$. In
 122 contrast, the niche-expert structure gives higher payoff to individuals who have both
 123 better predictive accuracy and a minority interest: $\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2$. Here, ρ_{e_A}
 124 denotes the proportion of players interested in the same factor, thereby allowing the
 125 niche-expert structure to preserve interest diversity. We note collective intelligence
 126 fails to emerge under the combination of averaging aggregation with niche-expert
 127 structure in the absence of AI (Wang et al., 2025).

128 Through time, the group evolves by allowing each individual to imitate peers
 129 who received higher payoffs (Figure 1C). Specifically, after each generation, two
 130 individuals are randomly sampled, and one imitates all three strategies (the factor of
 131 interest, factor belief, and reliance on AI) of the other with a probability determined
 132 by the difference in their expected payoffs (Materials and Methods). Individual

133 predictions are aggregated into a collective prediction $\hat{Y}$ based on the aggregation
 134 rules. Collective accuracy is defined as the reduction in prediction error achieved
 135 by the collective prediction relative to a baseline prediction α_0 (Wang et al., 2025):
 136 $1 - \mathbb{E}[(Y - \hat{Y}^{\text{re}})^2] / \mathbb{E}[(Y - \alpha_0)^2]$. Hereafter, we say that CI emerges when the collective
 137 accuracy reaches a near perfect level.

138 This model now allows us to explore the impact of AI on CI depending on the
 139 type of AI and the incentive structure. Here, we begin by comparing the emergence
 140 of CI with and without AI and subsequently investigate a how different incentive
 141 structures affect the emergence of CI in the presence of AI.

142 AI can either promote or prevent the emergence of 143 CI

144 To understand the impact of AI, we first compare collective dynamics with and
 145 without AI. First, consistent with Wang et al. (2025), simple imitation rules and
 146 incentive structures allow collective intelligence to emerge in the absence of AI (Fig.
 147 2A,C,D). One exception is the use of simple averaging with niche-expert incentive
 148 structures, under which the collective accuracy of the population remains near zero.
 149 This is because enhancing individual accuracy causes individuals to predict close to
 150 the true coefficient of their interest, but simply averaging these predictions does not
 151 necessarily produce an accurate estimate of the full prediction target (Fig. S1).

152 We find that AI usage can have heterogeneous outcomes depending on its type.
 153 For example, the use of Omniscient AI can either preserve CI under the feedback
 154 incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-
 155 expert incentive structure (Fig. 2B). These results are robust across different levels
 156 of AI accuracy and can be shown analytically by assuming a sufficiently large group
 157 of players (Fig. S2; see SI Evolutionary dynamics section). **[SWP: need to get**
 158 **back.]**

159 In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting
 160 in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive
 161 structure, the use of Chatbot AI leads to a loss of diversity in the population's
 162 interest, making it impossible to account for effect of some causal factors (Fig. S2,
 163 S3). This causes an increase in collective variance and stochastic decrease in collective
 164 accuracy (Fig. 2C; Fig. S3). We note that the loss of interest diversity may arise
 165 even in the absence of AI in finite populations (Fig. S3), but the use of Chatbot
 166 AI makes the group more vulnerable to such loss (Fig. S4). Under the niche-
 167 expert incentive structure, the population becomes overly reliant on AI, causing the
 168 collective prediction to inherit the AI's bias (Fig. S3). This leads to an increase in
 169 collective bias and a decrease in collective accuracy (Fig. 2; Fig. S3). Interestingly,
 170 we find that the niche-expert structure can prevent the use of AI under specific
 171 assumptions about the relationship between the true coefficient and AI bias (Fig.
 172 S2). In this case, the population reduces **[JS: I think it is difficult to say that it**

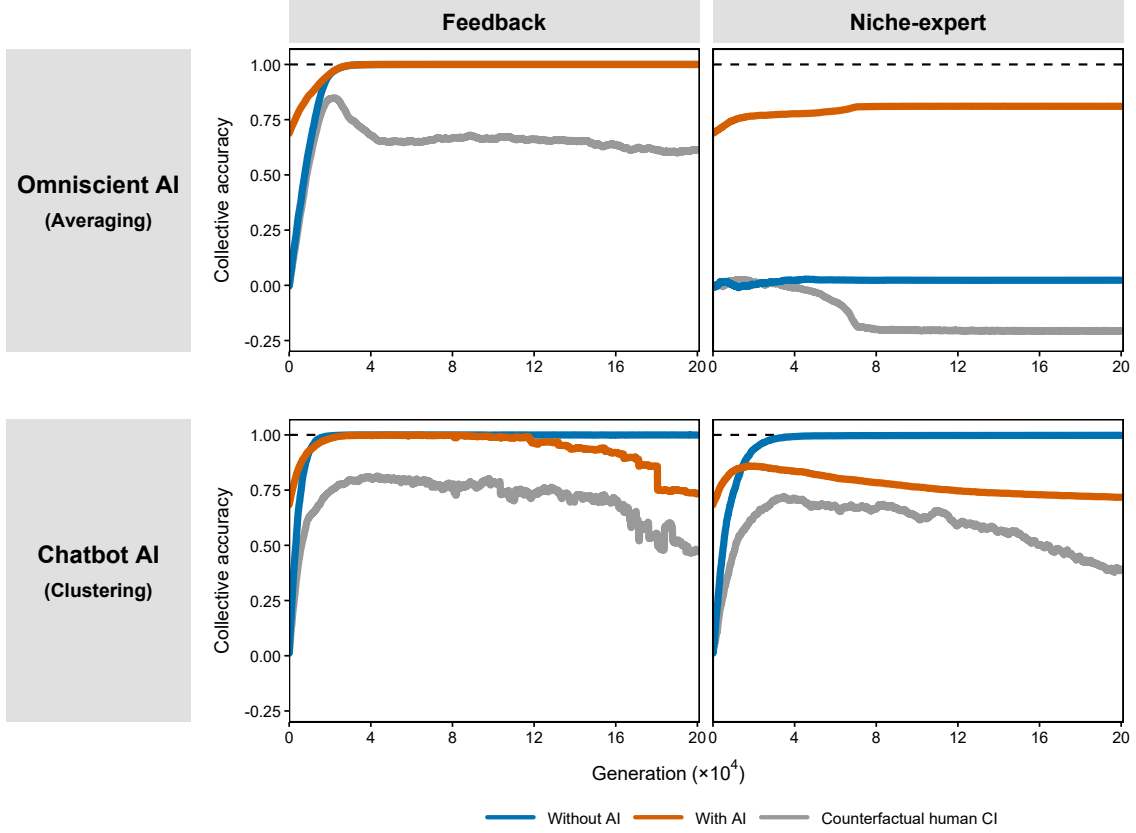


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Blue lines represent trajectories of collective accuracy without AI assistance. Orange lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players’ unrevised predictions, indicating the group’s predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

completely excludes AI, because reliance on AI does not approach zero. This may be due to the limited population size.] the reliance on AI assistance, allowing the CI to emerge eventually. However, this phenomenon can substantially delay the emergence of CI (Fig. S5).

To investigate whether the group can retain its predictive ability when AI assistance suddenly becomes unavailable at any given time, we quantify counterfactual collective accuracy based on unrevised predictions. Across all scenarios, we find that

the counterfactual collective accuracy is lower than the collective accuracy of the corresponding population (Fig. 2).

In summary, an omniscient AI can promote CI, but AI operating with limited information may prevent the emergence of CI. Moreover, reliance on AI can limit inherent ability of the population to make accurate predictions on their own, raising concerns about future AI usage. These findings further underscore the need for incentive structures that can mitigate the potential risks associated with AI assistance.

Directly incentivizing or penalizing AI use can additionally harm CI

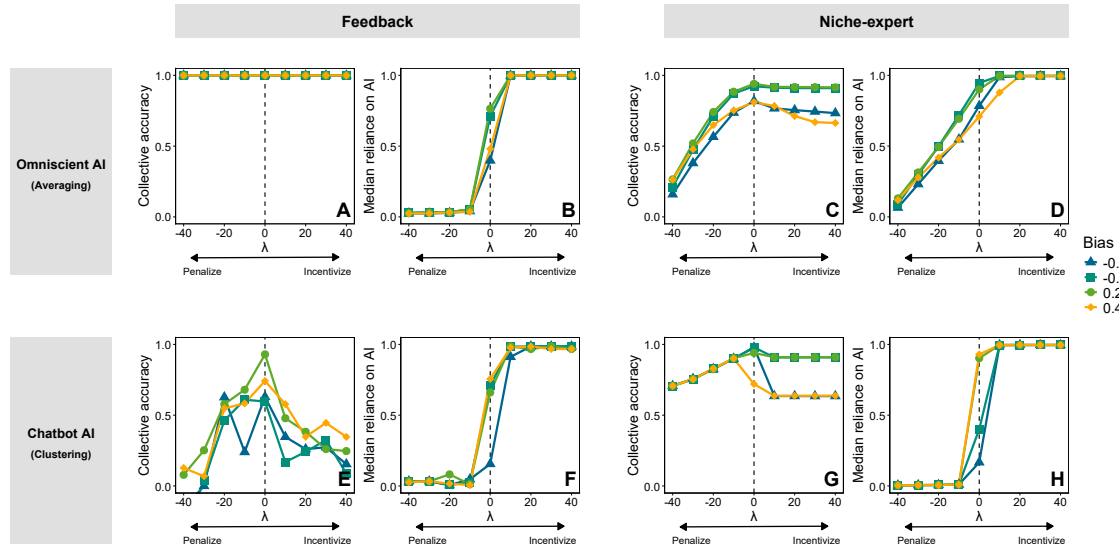


Figure 3: Collective accuracy under the influence of direct intervention on AI use. Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different assumptions about AI types (omniscient AI, A–D and chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

Given the strong influence of AI on collective outcomes, a straightforward inter-

vention is to directly incentivize or penalize the use of AI. To investigate whether such direct incentives or penalties can help the emergence of CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

where λ represent the strength of intervention ($\lambda > 0$) or penalty ($\lambda < 0$), and β_A represent individual reliance on AI. [SWP: maybe we need to call incentive pi earlier on...] [SWP: also good to define beta again here as a reminder.]

Directly incentivizing or penalizing AI use can further reduce the collective accuracy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure; in the latter case, the population retains collective intelligence regardless of the intervention strength (Fig. 3A). The reduction in collective accuracy arises because incentivizing or penalizing AI usage causes individuals to imitate others based on their AI usage rather than their predictive accuracy. Thus, stronger interventions can push the group to always [JS: wholly?] ($\beta = 1$) or never ($\beta = 0$) rely on AI (Fig. 3B, D, F, H) but these scenarios do not necessarily lead to better outcomes (Fig. 3D, F, H). When AI bias is relatively large, a weak penalty can lead to higher collective accuracy, but this represents a narrow parameter regime (Figure 3G).

Overall, these results indicate that simply incentivizing or penalizing the AI, independent of individuals' performance, use may exacerbate rather than resolve the risk of AI on CI. Thus, we also considered an incentive structure where interventions for AI usage affects the payoffs multiplicatively, thus giving higher payoffs to individuals who rely more (or less) AI and makes better prediction. [SWP: Supp.] [JS: Should I add the multiplicative intervention to the SI? I remember that we didn't reach a clear conclusion during our last meeting about whether to include it.]

Balanced incentives mitigate AI-induced failures of CI

Neither the original incentive structures nor simple interventions targeting AI use are sufficient to promote CI. What incentive structure, then, would work in the presence of AI? Here, we propose a "balanced" incentive structure, modeled as the average of the feedback and niche expert incentives.

The core idea behind the balanced structure is to maintain interest diversity and prevent collective bias. On one hand, the feedback incentive structure is prone to the loss of interest diversity but can prevent collective bias by incentivizing players whose predictions lie opposite to the collective prediction. On the other hand, the niche-expert structure can maintain interest diversity because it directly incentivizes minority interests. However the niche-expert structure cannot prevent collective bias because, within the same factor of interest, individuals are rewarded only according to their individual predictive performance, which in turn homogenize the predictions.

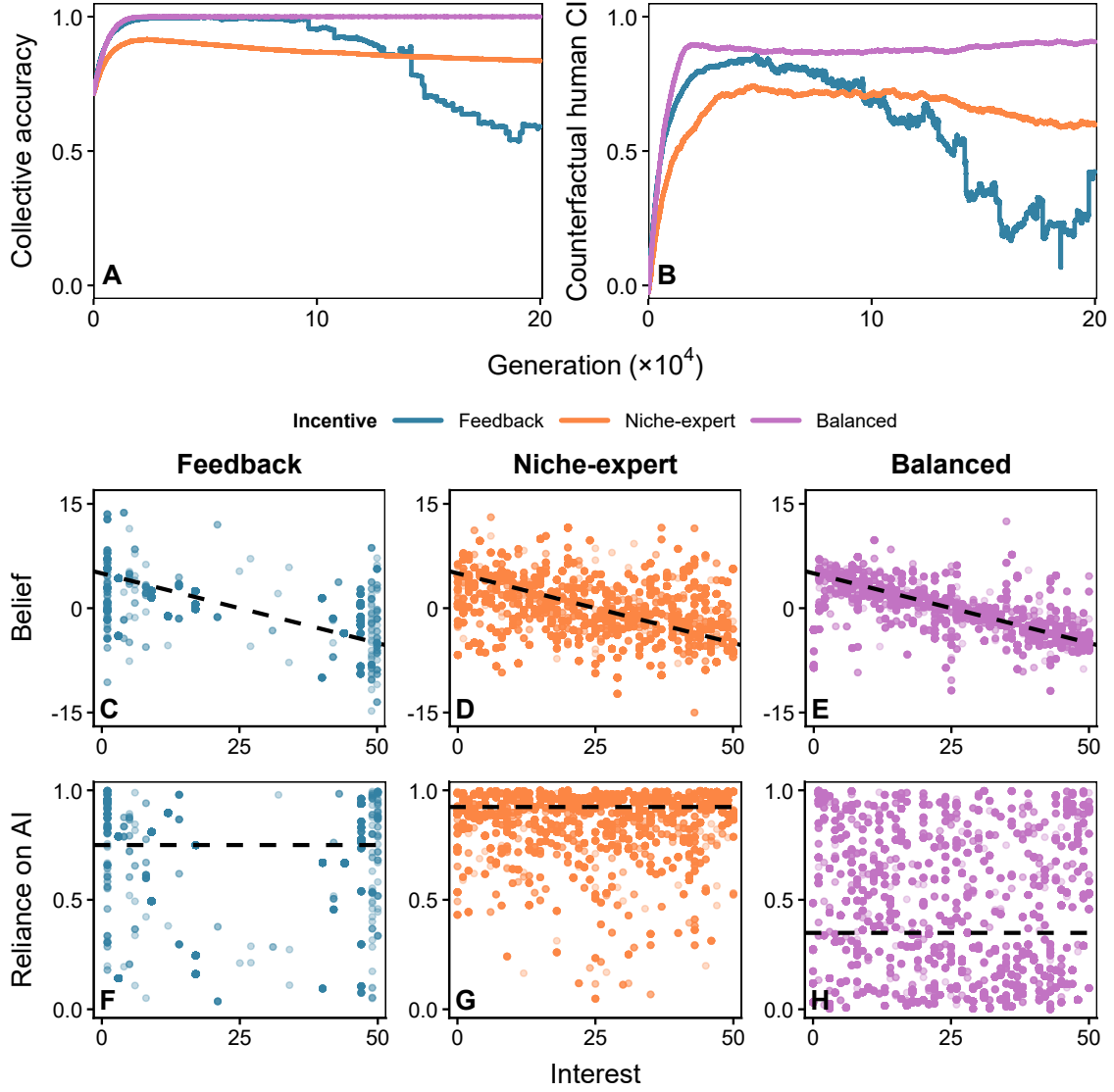


Figure 4: **Balanced structure can promote collective intelligence** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

Therefore, when the two incentive structures are combined, the population achieves CI by maintaining interest diversity while limiting collective bias (Fig. 4A, S6). Un-

der the balanced structure, individuals remain more evenly distributed across causal factors, and their beliefs lie closer to the true coefficients than under either incentive structure alone (Fig. 4C–E). The balanced structure also produces higher counterfactual collective accuracy than the feedback or niche-expert structure (Fig. 4B, S6). These patterns are robust to how the two incentive structures are weighted when combined (Fig. S7), and are driven by the emergent property of the balanced structure, under which the population increases its reliance on AI as AI becomes more accurate (Fig S8).

[SWP: *may need some robustness check rather than a case study...*]

Discussion

The ongoing race towards the development of “better” AIs continues to pose questions about the societal benefits and risks of AI uses (Bender et al., 2021; Burton et al., 2024; Cui and Yasseri, 2024). Here, we used mathematical models of evolutionary dynamics to investigate how AI uses and incentive structures jointly affect the emergence of CI, which is defined as the population’s ability to make accurate predictions. We find that an “omniscient” AI model, which incorporates information about all relevant causal factors, can improve collective accuracy, whereas a “chatbot” AI model, whose outcome depends on the user’s interest, can impair it. These contrasting outcomes arise because AI assistance affects not only individual predictions, but also the population’s interest diversity, beliefs, and reliance on AI. We also show that directly incentivizing or penalizing AI uses, independent of individual’s performances, can further limit the collective accuracy of the population. Instead, a balanced incentive structure can maintain CI by promoting diversity and limiting bias.

Our analysis reveals that AI can prevent the emergence of CI through two different mechanisms. First, AI use can cause individuals to concentrate on a subset of the factors that determine the prediction target, thereby reducing diversity in their interests. This result is broadly consistent with recent empirical studies showing that AI assistance can improve individual performance but limit the diversity of outputs, in domains ranging from scientific research (Hao et al., 2026) to creative writing (Doshi and Hauser, 2024). Second, over-reliance on AI can cause the population to inherit inherent biases in AI’s predictions (Fügener et al., 2021; Vicente and Matute, 2023), causing the prediction accuracy to decrease eventually. This mechanism can prevent the emergence of CI even when the interest diversity is maintained because biases in AI predictions can be amplified when individual predictions are aggregated. Thus, preserving the diversity of ideas will not be sufficient to promote CI. Instead, AI uses need to be calibrated to prevent systematic AI biases from becoming collective biases (Buçinca et al., 2021).

Our analyses of counterfactual collective accuracy also raises an important concern for future AI uses: while AI uses may improve the apparent performance of

individuals and, in some cases, the population as a whole, it can simultaneously reduce the population’s ability to make accurate predictions without AI. These results echo findings from a recent field experiment, which demonstrated that students, who seemingly performed better in solving mathematics problems under AI assistance, performed worse on their own than those who did not have access (Bastani et al., 2025). These findings highlight the need to evaluate AI tools not only by the performance gains they provide during use, but also by whether individuals and populations retain the ability to perform without them (Melumad and Yun, 2025; Ke et al., 2026). Making this distinction clear will be particularly important for education, where AI may help students complete a task without helping them learn how to do it on their own (Bastani et al., 2025; Melumad and Yun, 2025).

As generative AIs and large language models are already generating substantial benefits, many organizations, both industrial and academic, are increasingly seeking to improve their collective performance by regulating AI uses (Thorp, 2023; Bousquette, 2026; Chawla, 2026; Keary, 2026). Such regulations may take direct forms, including AI-use policies or token-based leaderboards, or may operate indirectly through other incentives, such as productivity rewards or the social stigma associated with AI use (Reif et al., 2025). Our model suggests that directly incentivizing or penalizing AI use is unlikely to induce appropriately calibrated reliance on AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups to propagate AI biases, whereas penalizing AI use could lead to underreliance, preventing groups from using AI even when it performs well in the relevant domain. Our result instead shows that appropriate AI use and accurate CI can be promoted by focusing on the traditional properties of groups that have been emphasized for decades: maintaining diversity and preventing collective bias, rather than on AI use itself (Hong and Page, 2004; Lorenz et al., 2011; Mann and Helbing, 2017).

There are several limitations to our analyses. First, our model allows for social learning among humans, but AI’s inherent ability remains fixed throughout the evolutionary process. In reality, AI systems can also learn from human-generated data (Ouyang et al., 2022). Such bidirectional learning increase the similarity between human and AI errors and therefore further reduce the group performance (?). Second, we model AI errors using fixed, factor-specific additive biases and independent random errors. This assumption allows random errors to be reduced through aggregation, while systematic biases can still persist in the collective prediction. In reality, AI errors may have more complex structures, including biased estimates of factor effects and correlated errors across individuals, factors, or repeated interactions. Further empirical and theoretical work is therefore needed to understand how these error structures affect the emergence of CI. Finally, following established models of CI, we assume that the prediction target is a linear combination of independent causal factors. This assumption does not hold for many real-world tasks, where causal factors may be correlated or interact nonlinearly. Future studies should therefore investigate how nonlinear relationships between causal factors and prediction targets affect the interaction between AI assistance and the emergence of CI.

313 The rapid adoption of personalized AI assistants is changing how individuals
 314 learn, work, and make decisions (Tessler et al., 2024; Glickman and Sharot, 2025;
 315 Sajadieh et al., 2026; Bick et al., 2026; Ke et al., 2026). While the immediate ben-
 316 efits and risks of these systems have been evaluated at the individual level Noy and
 317 Zhang (2023); Bastani et al. (2025), our study raises a broader concern that their
 318 widespread AI use may reshape collective intelligence of human society. Our results
 319 further underline the importance of designing reward structures that preserve minor-
 320 ity expertise and prevent AI biases from persisting across the population. A detailed
 321 empirical understanding of how AI assistance reshapes diversity, expertise, and re-
 322 liance at the population level will therefore be critical for predicting the dynamics of
 323 future of human collective intelligence.

324 Methods

325 Prediction target

326 Following previous models of CI, we model the prediction target as a linear combi-
 327 nation of M independent causal factors (Hong et al., 2012; Mann and Helbing, 2017;
 328 Wang et al., 2025):

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

329 where the coefficient α_i determines the magnitude and direction of the effect of
 330 factor i , and α_0 represents a global intercept. Each causal factor X_i is modeled as an
 331 independent Gaussian random variable with mean zero and standard deviation σ_i .

332 Players

333 We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel
 334 prediction task. In the absence of AI assistance, the strategy of player A is char-
 335 acterized by two variables: interest e_A and belief c_A . Player A observes the causal
 336 factor X_{e_A} and makes a prediction by multiplying the observed factor value by their
 337 belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

338 where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does
 339 not observe any causal factor and instead makes a constant prediction based solely
 340 on their belief. We refer to this as a stubborn strategy because the player’s predic-
 341 tion does not respond to the realized values of the causal factors. For notational
 342 convenience, we assume that $X_0 = 1$.

343 When the group is assisted by AI, each player’s strategy additionally includes
 344 reliance on AI, denoted by β_A . We assume that each player consults the AI inde-
 345 pendently and revises their prediction by taking a weighted average of their own
 346 prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor i is denoted by b_i , and the stochastic error is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. For simplicity, we assume $b_i = b$ for the main analysis for all i . The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation τ . Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore hold τ constant and vary the magnitude of the additive bias to manipulate the level of AI accuracy during our analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all M causal factors and incorporate them into its prediction regardless of the user's interest. The prediction generated by the Omniscient AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

The second is termed the Chatbot AI, which can use only the single causal factor X_{e_A} associated with the player's interest. The prediction generated by the Chatbot AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous research (Wang et al., 2025), depending on the type of AI used by the group. First, when the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all M factors, even though the individual's own prediction is based on only one factor. Each revised prediction can therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{N}. \quad (7)$$

382 Second, when the group is assisted by the Chatbot AI, both the player’s initial
 383 prediction and the AI’s prediction are based on the same causal factor. Each revised
 384 prediction therefore contains information about only one factor and can be treated
 385 as an estimate of that factor’s contribution to the target. We consequently use
 386 clustering aggregation.

387 Clustering aggregation is performed by grouping individuals according to their
 388 interests and calculating the mean prediction within each group. The collective
 389 prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

390 where $\mathcal{A}_i$ represents the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ represents
 391 the number of players in that set.

392 When the group is not assisted by AI, collective predictions are computed from the
 393 unrevised individual predictions throughout the simulation. Both simple averaging
 394 and clustering can be used in the absence of AI. For fair comparison, we use the
 395 same aggregation rule as in the corresponding AI-assisted condition.

396 Incentive structures

397 Players receive payoffs based on the underlying incentive structure. We consider five
 398 incentive structures: feedback structure, niche-expert structure, direct intervention
 399 on AI use of these two structures, and a balanced structure.

400 The feedback structure does not directly reward individuals according to their
 401 own prediction accuracy. Instead, it rewards predictions that are aligned with the
 402 residual error of the collective prediction and therefore have the potential to move
 403 the collective prediction toward the true value. Given player A ’s revised prediction
 404 Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

405 The niche-expert structure assigns higher payoffs to individuals who make accu-
 406 rate predictions while having an underrepresented interest. Specifically, the squared
 407 prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players
 408 interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

409 Previous research has shown that these two incentive structures can produce near
 410 perfect collective accuracy through social learning, except when the niche-expert
 411 incentive is combined with simple averaging (Wang et al., 2025). They are therefore
 412 used as baseline structures for examining how AI assistance affects the emergence of
 413 CI.

414 To examine the effects of directly encouraging or discouraging AI use, we mod-
 415 ify the feedback and niche-expert structures by adding a term proportional to the
 416 player’s reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

417 Here, λ represents the strength of the direct incentive or penalty applied to AI use. A
 418 positive value of λ rewards AI reliance, whereas a negative value penalizes it. When
 419 $\lambda = 0$, no direct incentive or penalty is applied, and the payoff functions reduce to
 420 the original feedback and niche-expert structures.

421 The balanced structure is defined as the arithmetic mean of the feedback and
 422 niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

423 Simulations

424 All simulations are conducted using an agent-based model with 50 causal factors
 425 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 426 target are sampled at the beginning of the simulations and subsequently held fixed.

427 Coefficients α_i are sampled from a uniform distribution on $[-5, 5]$, except in the
 428 simulation presented in Figure 4, in which α_i is set to $5 - 0.2i$ for visualization.
 429 Standard deviations σ_i are sampled from a uniform distribution on $[0, 3]$.

430 Players’ initial strategies are also sampled from predefined probability distribu-
 431 tions. The distribution of individual belief c_A depends on the aggregation rule. Under
 432 simple averaging, each factor’s contribution is scaled by the proportion of players in-
 433 terested in that factor. We therefore sample beliefs from a Gaussian distribution
 434 with mean zero and standard deviation 100, following the scaling used in previous
 435 work (Wang et al., 2025). Under clustering aggregation, beliefs are sampled from a
 436 Gaussian distribution with mean zero and standard deviation 5. Because predictions
 437 are averaged separately within each interest group and then summed, no additional
 438 scaling of c_A is required. The initial interest of each player is sampled uniformly
 439 from the integers between 0 and 50, and reliance on AI was sampled from a uniform
 440 distribution on $[0, 1]$.

441 We assume that prediction events occur on a substantially faster timescale than
 442 social learning. In each generation, we therefore calculate the expected payoff $\mathbb{E}[\pi_A]$
 443 of each player analytically rather than explicitly sampling prediction events. The
 444 expectation is taken over the distributions of the causal factors $X_1, \dots, X_M$ and the
 445 stochastic AI error ϵ , conditional on the current population strategies and the fixed
 446 target parameters. Closed form expression of expected payoffs are derived for all
 447 incentive structures and are presented in the Supplementary Methods.

Each generation consists of an expected payoff calculation followed by one opportunity for pairwise social learning. Two distinct players, A and B , are randomly sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

where s represents the strength of social learning and was set to 50 in all simulations. We introduce a mutation rate μ , such that with probability μ , player A does not imitate player B but instead redrew all strategy components from the same distributions used to initialize the population. The mutation rate is set to zero in all simulations except those presented in Supplementary Fig. S4 [SWP: *need to check*].

Unless otherwise specified, each simulation is run for 200,000 generations. When a result is represented by a single summary value rather than as a temporal trajectory, the metric was averaged over the final 10,000 generations.

To evaluate the dynamics of collective prediction, we define six summary metrics: collective accuracy, collective bias, collective variance, counterfactual collective accuracy, interest diversity, and median reliance on AI. At each population state, expectations and variances are evaluated analytically over the distributions of the causal factors and AI errors rather than estimated from sampled prediction events. Collective bias and collective variance constitute the bias–variance decomposition of the group level collective prediction error.

Collective accuracy is normalized relative to the baseline prediction α_0 . A negative value indicates that the collective prediction performs worse than the baseline. Counterfactual collective accuracy is evaluated by setting $\beta_A = 0$ while retaining the evolved interests and beliefs of the AI-assisted population. In calculating interest diversity, we use the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

Data and code availability

All data are stored in a publicly available GitHub repository (<https://github.com/parklab-snu/collective-intelligence-with-AI>).

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Contributions

Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology, Software, Visualization, Writing - Original Draft Preparation, Writing - Review &

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

481 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
482 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

483 **Competing Interests**

484 The authors declare no competing interests.

485 Supplementary Text

486 Here, we derive the expected payoffs and collective accuracy used in the simulations.
487 We consider Omniscient AI and Chatbot AI separately because they contain differ-
488 ent amounts of information and therefore require different aggregation rules. We
489 then analyze the corresponding evolutionary dynamics under a deterministic large-
490 population approximation.

491 1 Omniscient AI

492 We first consider Omniscient AI, which incorporates all causal factors regardless of
493 the player's interest. We derive individual and collective predictions before calculat-
494 ing the expected payoffs under each incentive structure.

495 1.1 Model

496 Because Omniscient AI observes all M causal factors, its prediction for player A is
497 given by

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon, \quad (\text{S1})$$

498 where b_i represent factor-specific biases and ϵ represents stochastic error, following
499 a Gaussian distribution with mean zero and standard deviation τ . A new stochastic
500 error is generated independently for each player and prediction event.

501 Let $B = \sum_{i=0}^M b_i$ denote the total additive bias. Because the true prediction
502 target is $Y = \alpha_0 + \sum_{i=1}^M \alpha_i X_i$, the AI prediction can be written as

$$Y^{AI} = Y + B + \epsilon. \quad (\text{S2})$$

503 In contrast to the AI prediction, player A 's initial prediction depends only on the
504 causal factor associated with their interest. Their initial and revised predictions are
505 therefore given by:

$$Y_A = c_A X_{e_A} \quad (\text{S3})$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (\text{S4})$$

506 represents player A 's interest, c_A represents their belief about the effect of the corre-
507 sponding factor, and β_A represents their reliance on AI. As in the main text, we set
508 $X_0 = 1$, such that a player with $e_A = 0$ makes a constant initial prediction.

509 1.2 Aggregation

510 Because the AI component of each revised prediction contains information about all
511 M causal factors, we use simple averaging to obtain the collective prediction:

$$\hat{Y}^{re} = \frac{1}{N} \sum_{j=1}^N Y_j^{re} \quad (S5)$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j Y^{AI} + (1 - \beta_j) Y_j\} \quad (S6)$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j}\} \quad (S7)$$

512 To separate the contributions of AI and the players' initial predictions, we define the
513 population's mean reliance on AI, the stochastic AI error remaining after aggregation,
514 and the aggregate contribution of initial predictions associated with each factor:

$$\bar{\beta} = \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S8)$$

$$\bar{\eta} = \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (S9)$$

$$\gamma_i = \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (S10)$$

515 Here, $\bar{\beta}$ represents the mean reliance on AI, $\bar{\eta}$ represents the stochastic AI error
516 remaining after averaging, and γ_i represents the contribution of players' initial pre-
517 dictions to factor i . Using these definitions, the revised collective prediction can be
518 written as follows:

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (S11)$$

519 Thus, the collective prediction consists of an AI-derived component, the contributions
520 of the players' initial predictions, and the stochastic AI error that remains after
521 aggregation.

522 1.3 Expected payoffs

523 We next derive the expected payoff under each incentive structure. Expectations are
524 taken over the distributions of the causal factors and stochastic AI errors, conditional
525 on the current population strategies and the fixed parameters of the prediction target.

526 Feedback

527 Under the feedback structure, a player's payoff depends on the alignment between
 528 their revised prediction and the residual error of the collective prediction. To express
 529 this residual, we first define the remaining error in the intercept and in the coefficient
 530 of each causal factor:

$$\delta_0 = \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0), \quad (\text{S12})$$

$$\delta_i = \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0), \quad (\text{S13})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S14})$$

531 where δ_0 represents the remaining error in the intercept, whereas δ_i represents the
 532 remaining error associated with factor i . Because the causal factors and stochastic AI
 533 errors are independent and have zero mean, cross terms vanish when the expectation
 534 is taken. Substituting the revised individual prediction and collective residual into
 535 the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S15})$$

$$= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \right] \quad (\text{S16})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \quad (\text{S17})$$

$$= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \quad (\text{S18})$$

$$= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \quad (\text{S19})$$

536 The first term represents the alignment between the AI-derived prediction and the
 537 collective residual, whereas the second term represents the contribution of the player's
 538 initial prediction. The final term arises because the stochastic error in player A 's AI
 539 prediction also contributes to the collective prediction. This finite-population effect
 540 decreases as population size increases.

541 Niche-expert

542 Under the niche-expert structure, the payoff is the negative squared error of the
 543 revised individual prediction, weighted by the frequency of the player's interest.
 544 Because $X_0 = 1$, a player with $e_A = 0$ makes a constant initial prediction. In this

case, the expected payoff is:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S20})$$

$$= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \quad (\text{S21})$$

$$= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \quad (\text{S22})$$

$$= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \quad (\text{S23})$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S24})$$

For a player with $e_A \neq 0$, the initial prediction instead depends on the observed value of factor X_{e_A} . The expected payoff is therefore:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S25})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \quad (\text{S26})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)\alpha_0 + \beta_A B$$

$$+ (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\}^2] \quad (\text{S28})$$

$$= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B\right. \quad (\text{S29})$$

$$+ \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S31})$$

These expressions separate the contributions of the player's belief, the unobserved causal factors, systematic AI bias, and stochastic AI error. Increasing reliance on AI reduces the contribution of errors in the player's initial prediction but increases their exposure to AI bias and stochastic error.

Incentivizing or penalizing AI use

Direct incentives add the term $\lambda\beta_A$ to either the feedback or niche-expert payoff. Because this term is fixed conditional on the player's strategy, its expectation is simply $\lambda\beta_A$. The expected payoffs therefore become

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re})\right] + \lambda\beta_A, \quad (\text{S32})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{re} - Y)^2\right] + \lambda\beta_A. \quad (\text{S33})$$

Thus, a positive value of λ increases the payoff of strategies with greater reliance on AI, whereas a negative value favors strategies with lower reliance.

558 1.4 Expected accuracy

559 Using the residual defined above, the mean squared error of the revised collective
560 prediction is:

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S34})$$

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S35})$$

561 The expected collective accuracy is therefore

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S36})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S37})$$

562 The collective prediction error consists of the squared error in the intercept, the re-
563 maining error associated with each causal factor, and the stochastic AI error remain-
564 ing after averaging across players. Counterfactual collective accuracy is obtained
565 from the same expression by setting $\beta_A = 0$ for all players while retaining their
566 evolved interests and beliefs.

567 2 Chatbot AI

568 Unlike Omniscient AI, Chatbot AI uses only the causal factor associated with each
569 player's interest. This causes the information contained in AI predictions to depend
570 directly on the distribution of interests in the population. Here, we derive the collec-
571 tive prediction, expected payoffs, and expected collective accuracy under this form
572 of AI assistance.

573 2.1 Model

574 The prediction generated by Chatbot AI for player A depends on the causal factor
575 associated with the player's interest:

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon, \quad (\text{S38})$$

576 where b_{e_A} represents factor-specific additive bias, and ϵ represents stochastic error.
577 The stochastic error follows a Gaussian distribution with mean zero and standard
578 deviation τ and is generated independently across players and prediction events. The

579 player's initial and revised predictions are given by Player A's initial and revised
 580 predictions can be then written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S39})$$

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S40})$$

581 Here, c_A represents player A's belief about the effect of their focal factor, e_A repre-
 582 sents their interest, and β_A represents their reliance on AI. Thus, the revised predic-
 583 tion remains specific to the single factor associated with the player's interest.

584 2.2 Aggregation

585 Because both the player's initial prediction and the Chatbot AI prediction contain
 586 information about the same causal factor, we use clustering aggregation. Revised
 587 predictions are first averaged among players interested in the same factor and then
 588 summed across represented interests:

$$\hat{Y}^{re} = \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \quad (\text{S41})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \} \quad (\text{S42})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon], \quad (\text{S43})$$

589 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number of
 590 players in that set.

591 To simplify the collective prediction, we define the effective coefficient, weighted
 592 AI bias, and stochastic error associated with each interest group:

$$\tilde{c}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \}, \quad (\text{S44})$$

$$\tilde{B}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A}, \quad (\text{S45})$$

$$\bar{B} = \sum_{i=0}^M \tilde{B}_i, \quad (\text{S46})$$

$$\tilde{\eta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon, \quad (\text{S47})$$

$$\bar{\eta} = \sum_{i=0}^M \tilde{\eta}_i, \quad (\text{S48})$$

where $\tilde{c}_i$ represents the mean effective coefficient for factor i after AI assistance, whereas $\tilde{B}_i$ and $\tilde{\eta}_i$ represent its systematic and stochastic error components. Thus, the collective prediction can be written as:

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} \tilde{c}_i X_i + \bar{B} + \bar{\eta}, \quad (\text{S49})$$

where $\mathcal{I} = i \in 0, \dots, M > 0$ denotes the set of represented interests.

2.3 Expected payoffs

We next derive the expected payoff under each incentive structure. Expectations are taken over the causal factors and stochastic AI errors, conditional on the current population strategies and the fixed parameters of the prediction target.

Feedback

Under the feedback structure, players are rewarded according to whether their revised predictions reduce the error in the collective prediction. We therefore define the remaining errors in the intercept and factor-specific coefficients as

$$\delta_0 = \alpha_0 - \tilde{c}_0 - \bar{B}, \quad (\text{S50})$$

$$\delta_i = \alpha_i - \tilde{c}_i, \quad (i \neq 0), \quad (\text{S51})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S52})$$

where δ_0 includes the remaining error in the intercept and the total additive AI bias, whereas δ_i measures the remaining error in the coefficient of factor i . Because the causal factors and stochastic errors are independent and have zero mean, substituting the revised individual prediction into the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S53})$$

$$= \mathbb{E}[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \quad (\text{S54})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\}] \quad (\text{S55})$$

$$= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \quad (\text{S56})$$

Thus, the expected feedback payoff depends on the remaining collective error associated with the player's interest, the contribution of factor-specific AI bias to the intercept error, and the player's contribution to the stochastic error of their interest group.

613 Niche-expert

614 Under the niche-expert structure, the negative squared error of each revised indi-
 615 vidual prediction is weighted by the frequency of the player's interest. We consider
 616 $e_A = 0$ separately because these players make constant predictions rather than factor-
 617 dependent predictions. When $e_A = 0$, the expected payoff is

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S57})$$

$$= -\mathbb{E} \left[\rho_0 \left[\left\{ \beta_A \alpha_0 + (1 - \beta_A) c_A \right\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \quad (\text{S58})$$

$$= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0) \beta_A b_0 + \beta_A^2 b_0^2 \right. \quad (\text{S59})$$

$$\left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \quad (\text{S60})$$

618 When $e_A \neq 0$, the player's prediction varies with their focal causal factor. The
 619 expected payoff is then

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S61})$$

$$= -\mathbb{E} [\rho_{e_A} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} \quad (\text{S62})$$

$$+ \beta_A b_{e_A} + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i]^2] \quad (\text{S63})$$

$$= -\rho_{e_A} \left\{ (1 - \beta_A)^2 (c_A - \alpha_{e_A})^2 \sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 \right. \quad (\text{S64})$$

$$\left. + \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0 \beta_A b_{e_A} + \beta_A^2 \tau^2 \right\} \quad (\text{S65})$$

620 In both cases, the expected squared error consists of the mismatch between the
 621 player's belief and the true coefficient, contributions from causal factors omitted
 622 from the individual prediction, factor-specific AI bias, and stochastic AI error. The
 623 niche-expert weight ρ_{e_A} scales these errors according to the frequency of the player's
 624 interest.

625 Incentivizing or penalizing AI use

626 Directly incentivizing or penalizing AI use adds the term $\lambda \beta_A$ to the player's payoff.
 627 Because this term is fixed conditional on the player's strategy, it can be taken outside

the expectation. The expected payoffs are therefore

$$\mathbb{E}(\pi_{A,Feedback}) = \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \right] \quad (\text{S66})$$

$$= \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] + \lambda\beta_A, \quad (\text{S67})$$

$$\mathbb{E}(\pi_{A,Niche-expert}) = -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A \right] \quad (\text{S68})$$

$$= -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right] + \lambda\beta_A. \quad (\text{S69})$$

Balanced

Under the balanced structure, the payoff corresponds to the arithmetic mean of the feedback and niche-expert payoff components:

$$\mathbb{E}(\pi_A) = \mathbb{E} \left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2} \right] \quad (\text{S70})$$

$$= \frac{\mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] - \mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right]}{2} \quad (\text{S71})$$

The expected payoff under the balanced structure can therefore be calculated directly from the feedback and niche-expert payoffs derived above.

2.4 Expected accuracy

Finally, we derive the expected collective accuracy under clustering aggregation. The intercept α_0 serves as the baseline prediction. Because the causal factors are independent and have mean zero, its expected squared error is

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S72})$$

Using the expression for the collective residual above, the expected squared error of the revised collective prediction is

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S73})$$

The first two terms represent squared errors in the intercept and factor-specific coefficients. The final term represents the stochastic AI error remaining after predictions are averaged within each interest group. Thus, stochastic error has a larger effect when an interest is represented by fewer individuals. Normalizing this error relative

644 to the baseline gives:

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S74})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S75})$$

645 Here, $R = 1$ represents perfect collective prediction, $R = 0$ represents performance
646 equal to the baseline, and $R < 0$ represents performance worse than the baseline.

647 3 Evolutionary dynamics

648 For sufficiently large population ($N \rightarrow \infty$), we can approximate the deterministic
649 evolutionary dynamics using the replicator equation under both weak and strong
650 selections. We focus on the feedback and niche-expert structures, for which the long-
651 term dynamics can be characterized analytically. The direct interventions on AI use
652 and the balanced structure are examined through simulations.

653 Let λ denote a strategy, p_λ its frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff.
654 Then, the corresponding replicator equation can be written as:

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S76})$$

655 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$ represents the mean expected payoff of the population. At an
656 internal stationary state, all strategies with positive frequency must have the same
657 expected payoff. A boundary state can also be stationary because a strategy with zero
658 frequency cannot increase unless it is introduced into the population. We therefore
659 assess the stability of boundary states by asking whether they can be invaded by a
660 rare alternative strategy.

661 3.1 Omniscient AI

662 With Omniscient AI, every AI prediction contains information about the full predic-
663 tion target, and revised predictions are combined through averaging. As population
664 size increases, independent AI errors vanish from the collective prediction. How-
665 ever, these errors remain in individual prediction errors and can therefore continue
666 to affect niche-expert payoffs. We consider the two incentive structures separately.

667 3.1.1 Feedback

668 Under the feedback structure, both the stochastic error in the collective prediction,
669 $\bar{\eta}$, and its contribution to expected payoff vanish as $N \rightarrow \infty$. Following previous

work (Wang et al., 2025), we define the Lyapunov function as the expected error of collective prediction:

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S77})$$

The function V is non-negative and equals zero when the collective prediction is identical to the target. To determine how V changes over time, we take its time derivative:

$$\frac{dV}{dt} = -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \quad (\text{S78})$$

$$= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \quad (\text{S79})$$

$$= -2 \left[\left\{ (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right]. \quad (\text{S80})$$

For notational convenience, we define

$$K = (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S81})$$

Although K changes with the population state, it is common to all strategies at a given state and can therefore be treated as a constant with respect to the strategy index λ . We also define:

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S82})$$

Under the replicator equation, the rate of change in the population mean of a strategy component is given by its covariance with expected payoff.

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S83})$$

Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta)cG_e, u). \quad (\text{S84})$$

Therefore,

$$\frac{dV}{dt} = -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta)cG_e, u) \} \quad (\text{S85})$$

$$= -2 \text{Cov}(K\beta + (1 - \beta)cG_e, u). \quad (\text{S86})$$

For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy λ is

$$u_\lambda = K\beta_\lambda + (1 - \beta_\lambda)c_\lambda G_{e_\lambda}. \quad (\text{S87})$$

This expected payoff is identical to the quantity inside the covariance above. Therefore,

$$\frac{dV}{dt} = -2\text{Cov}(u, u) = -2\text{Var}(u) \leq 0. \quad (\text{S88})$$

Because payoff variance is non-negative, the expected collective prediction error cannot increase and decreases whenever the strategies currently represented in the population differ in their expected payoffs. The derivative becomes zero when all represented strategies receive the same expected payoff. However, this condition does not necessarily imply that $V = 0$, because a population can lack strategies that would improve its collective prediction.

If $V > 0$, at least one component of the collective residual remains nonzero. A rare strategy whose prediction is aligned with this residual can obtain a higher expected payoff than the resident strategies and invade the population. Thus, provided that novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$. At such a state, $\hat{Y}^{\text{re}} = Y$, and collective accuracy reaches $R = 1$.

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition arises when all represented strategies receive the same expected payoff. However, this condition does not necessarily imply that $V = 0$, because a population can lack strategies that would improve its collective prediction.

If $V > 0$, at least one component of the collective residual remains nonzero. A rare strategy whose prediction is aligned with this residual can obtain a higher expected payoff than the resident strategies and invade the population. Thus, provided that novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$. At such a state, $\hat{Y}^{\text{re}} = Y$, and collective accuracy reaches $R = 1$.

3.1.2 Niche-expert

Unlike feedback payoffs, niche-expert payoffs depend on the prediction error of each individual. Stochastic AI error therefore remains in the expected payoff even when $N \rightarrow \infty$, and we retain $\tau > 0$ throughout this analysis.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S89})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its

unconstrained stationary point is then:

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S90})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

To compare the optimized strategies across interests, let L_e denote the minimum expected individual squared error within interest e , $\mathbb{E}[(Y_A^{re} - Y)^2]$. For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S91})$$

If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating the expected squared prediction error at the corresponding boundary.

Interest 0 must be considered separately because $X_0 = 1$ and the player's belief determines a constant prediction. Optimizing with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S92})$$

Substituting this into the expected payoff gives a strictly concave function of β , with the unique optimum:

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S93})$$

Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S94})$$

and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S95})$$

At this optimum, the player adjusts their belief to offset the systematic AI bias. The stochastic component of AI error cannot be removed in the same way and therefore limits the benefit of AI reliance.

We denote the optimal strategy within interest e by S_e . When $\beta_e^* = 1$, different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class. A stable state can contain only the optimal strategy class within each represented interest because a suboptimal strategy can be invaded by S_e .

742 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
 743 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S96})$$

744 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator equa-
 745 tion at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S97})$$

746 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S98})$$

747 This internal stationary state is locally stable. At equilibrium, all represented in-
 748 terests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
 749 interest increases slightly above its equilibrium value, its payoff decreases because
 750 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of an
 751 interest decreases below its equilibrium value, its payoff increases. The replicator dy-
 752 namics therefore generate a restoring force that reduces small deviations in interest
 753 frequencies and drives the population back toward the internal stationary state.

754 If one or more interests are absent, the resulting boundary state can be stationary
 755 under the replicator equation, but it is unstable against invasion. The payoff of a rare
 756 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
 757 population is negative when $L_e > 0$, so the rare interest increases when introduced
 758 at sufficiently small frequency.

759 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
 760 prediction strategies and their frequencies. Whether the constrained optimum has
 761 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
 762 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
 763 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
 764 interest frequency remain uniquely determined.

765 Having characterized this equilibrium, we next compare its collective prediction
 766 with a prediction made entirely by AI. In particular, the expected squared error of
 767 an Omniscient AI prediction is

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S99})$$

768 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
 769 fore, because L_e is the minimum individual expected squared prediction error within
 770 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S100})$$

Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S101})$$

Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S102})$$

The collective prediction is the average of individual revised predictions. By convexity of the squared error,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S103})$$

Therefore,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] < \mathbb{E} \left[(Y - Y^{\text{AI}})^2 \right]. \quad (\text{S104})$$

Thus, the diversity of optimized predictions allows the population to outperform an individual AI prediction at equilibrium.

However, outperforming AI does not imply perfect collective accuracy. Under averaging, the non-AI contribution associated with a particular factor is provided only by individuals interested in that factor. The predictions of individuals with other interests therefore attenuate that factor's coefficient in the collective prediction. For a nonzero factor i , the residual coefficient of the stationary collective prediction can be written as

$$\delta_i^* = \alpha_i \left\{ 1 - \bar{\beta}^* - \rho_i^* (1 - \beta_i^*) \right\} = \alpha_i \sum_{e \neq i} \rho_e^* (1 - \beta_e^*). \quad (\text{S105})$$

At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore, for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S106})$$

Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error. Hence,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] > 0, \quad (\text{S107})$$

and consequently the stationary collective accuracy satisfies $R^* < 1$. Thus, the niche-expert structure preserves interest diversity and allows the population to outperform AI, but averaging prevents the population from reaching perfect collective accuracy.

Supplementary Figure

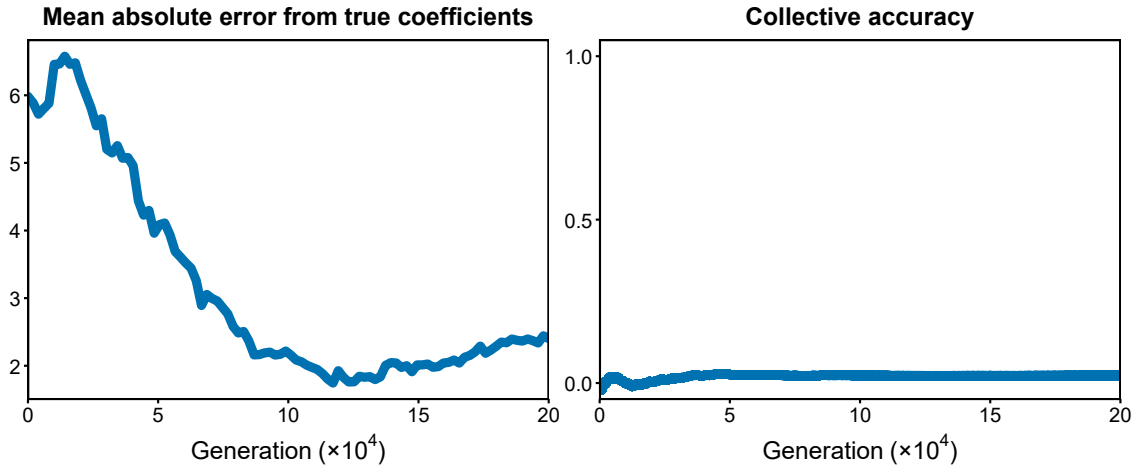


Figure S1: **Niche-expert structure with averaging aggregation cannot promote CI without AI.** Trajectories of the mean absolute error from true coefficients and collective accuracy under the niche-expert structure with averaging aggregation in the absence of AI. The mean absolute error from true coefficients is calculated by first taking, for each interest, the absolute difference between the true coefficient and the mean belief of individuals sharing that interest, and then averaging these differences across all $m + 1$ interests. The mean absolute error decreases across generations, indicating that the mean beliefs converge toward their corresponding true coefficients. However, collective accuracy remains near zero, indicating that simply averaging individual predictions does not yield accurate collective predictions. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$.

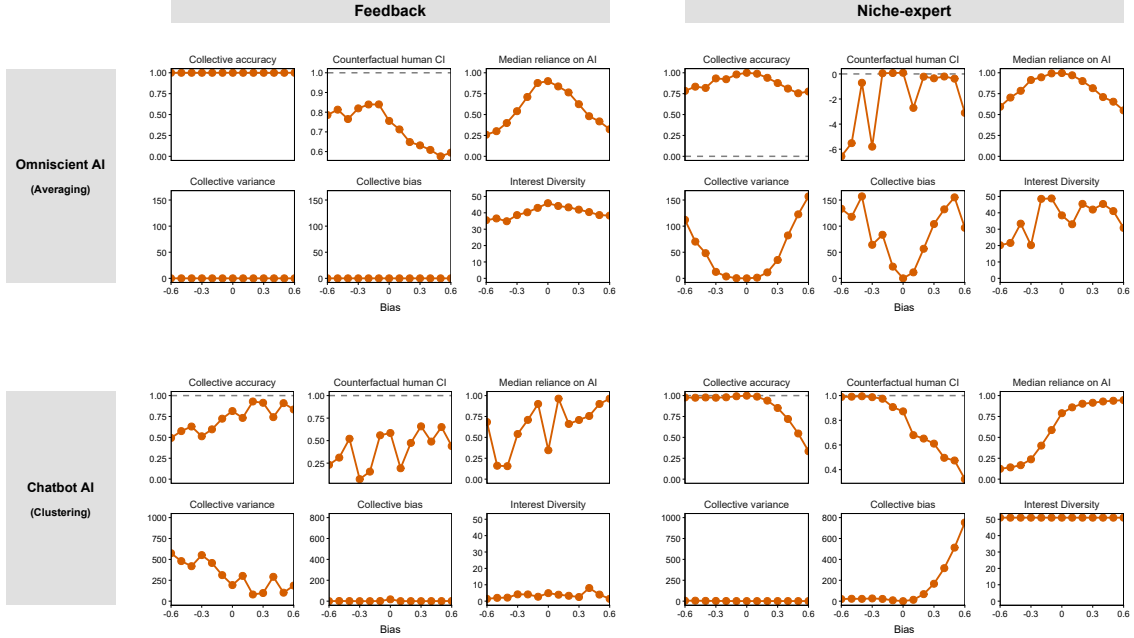


Figure S2: **Evaluation metrics across various levels of AI bias.** Evaluation metrics across varying levels of AI bias under different assumptions about AI types (omniscient AI, and chatbot AI) and incentive structures (feedback, niche-expert). Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. Dashed horizontal lines represent the values of the evaluation metrics when the group is not assisted by AI. Collective accuracy represents the predictive performance of the group, and counterfactual human CI represents the group’s predictive accuracy if AI assistance were unavailable at the given generation. Interest diversity is defined as Hill-Shannon diversity of interest within the population, median reliance on AI represents the median value of reliance on AI (β). Collective bias represents systematic deviation of the collective prediction from the true target, collective variance represents variability of the collective prediction error across prediction events. All the evaluation metrics are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$.

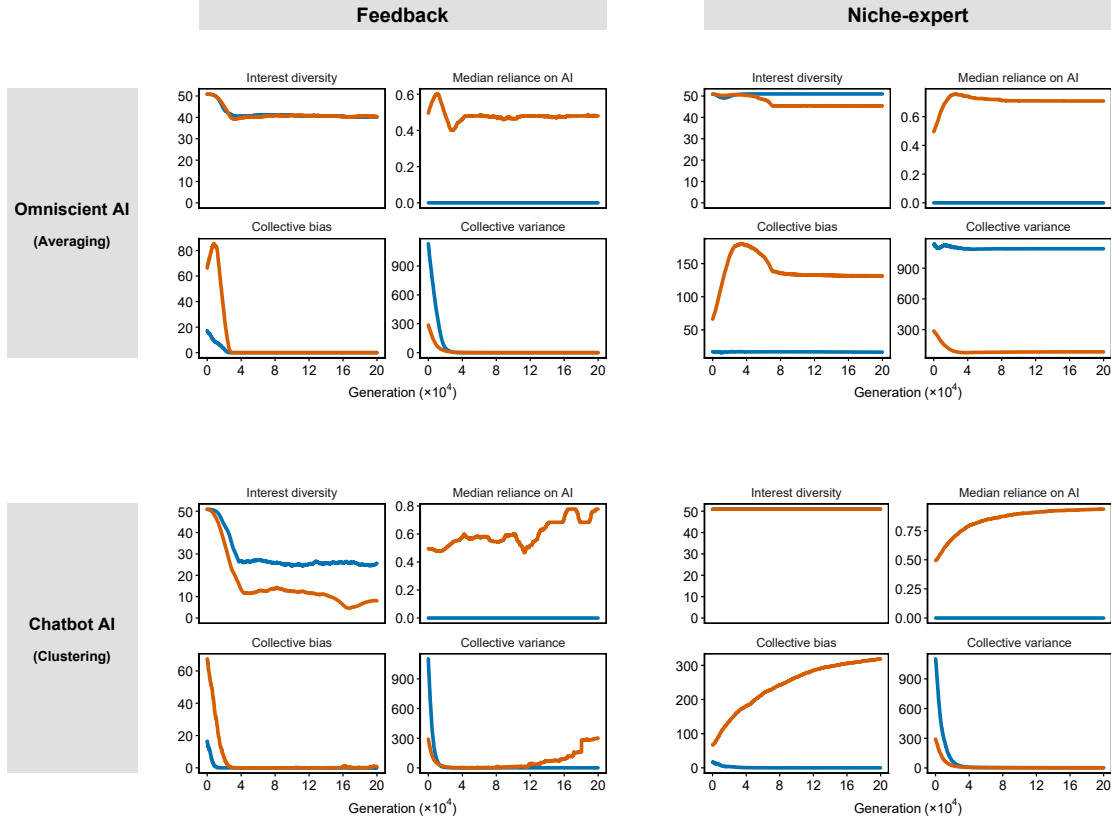


Figure S3: **Comparison of evaluation metrics with and without AI.** Trajectories of evaluation metrics when the group is assisted by the Omniscient AI and Chatbot AI. Here, feedback and niche-expert incentive structures are considered. Blue lines represent trajectories of evaluation metrics without AI assistance. Orange lines represent trajectories of evaluation metrics with AI assistance. Interest diversity is define as hill-shannon diversity of interest within population, median reliance on AI represent population median of reliance on AI (β), collective bias represent systematic deviation of the collective prediction from the true target, collective variance repressent variability of the collective prediction error across prediction events. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

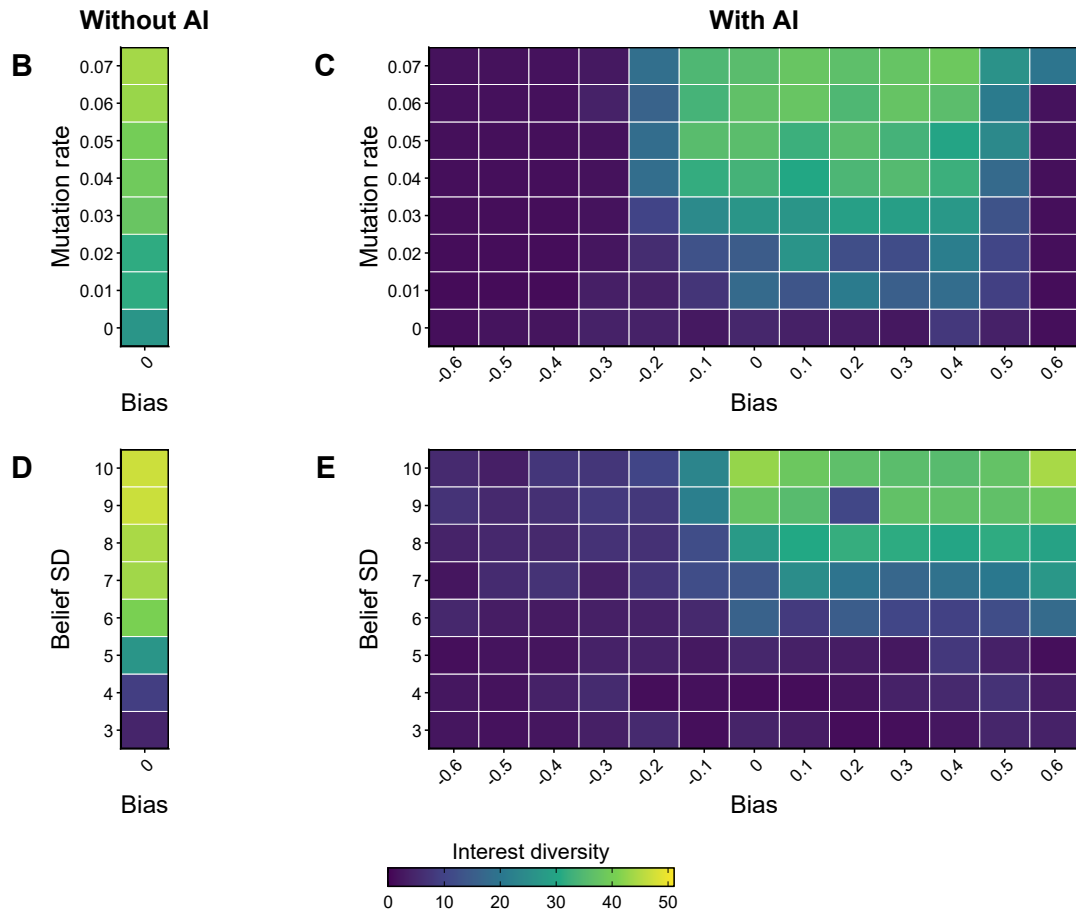
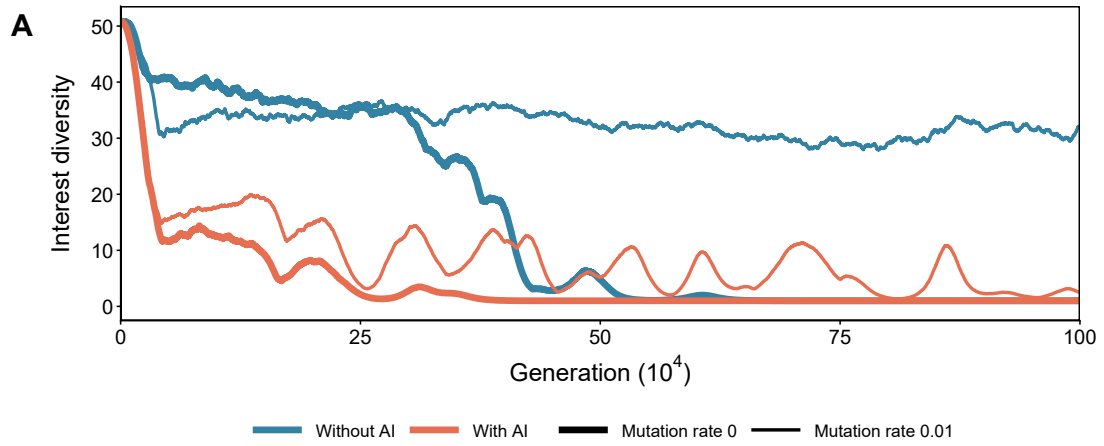


Figure S4: **Feedback structure becomes more prone to interest diversity loss with AI** Trajectories of interest diversity with and without AI under the feedback structure (A). Heatmaps of interest diversity across varying levels of mutation rate (B, C) and the standard deviation of the initial belief distribution (D, E), with and without AI. Interest diversity declines toward zero even without AI over long timescales when there is no mutation and the initial belief distribution is narrow. However, with AI, interest diversity declines more strongly, even under moderate mutation rates and wider initial belief distributions. Interest diversity in the heatmaps is averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$ for (D, E), $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$ for (A, B, C), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$, chatbot AI is used.

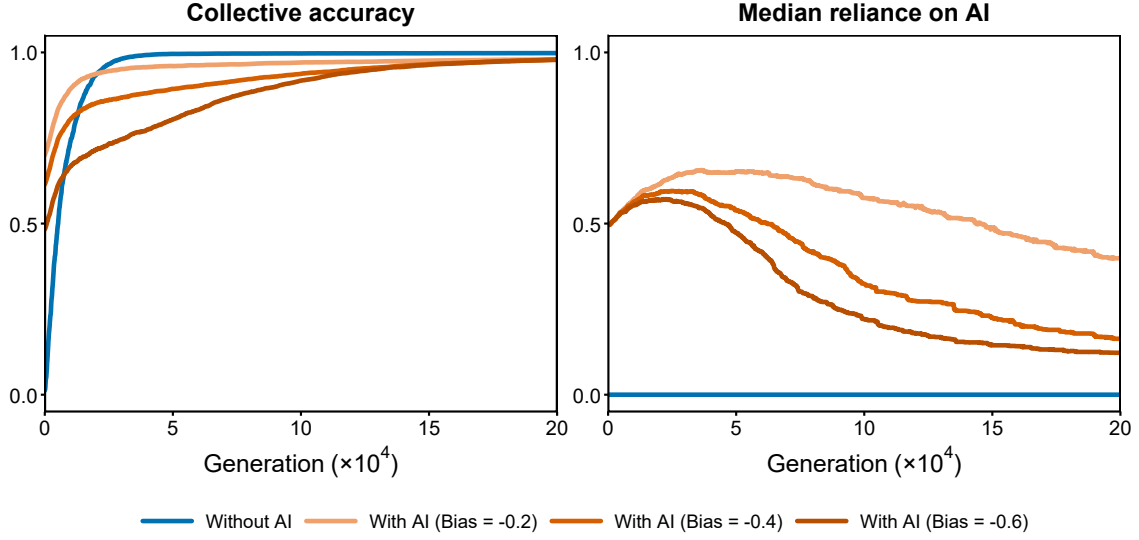


Figure S5: **Niche-expert structure may reduce reliance on AI, delaying the emergence of CI.** Trajectories of collective accuracy and median reliance on AI when the niche-expert structure causes reliance on AI to decline over time. Red lines represent trajectories with AI across varying levels of AI bias (-0.2, -0.4, and -0.6). Green lines represent trajectories without AI; reliance on AI is set to zero in the absence of AI. This reduction in reliance on AI occurs when the α_0 and AI bias have opposite signs, and is associated with the eventual emergence of CI, although CI emerges substantially later than in the condition without AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, chatbot AI is used.

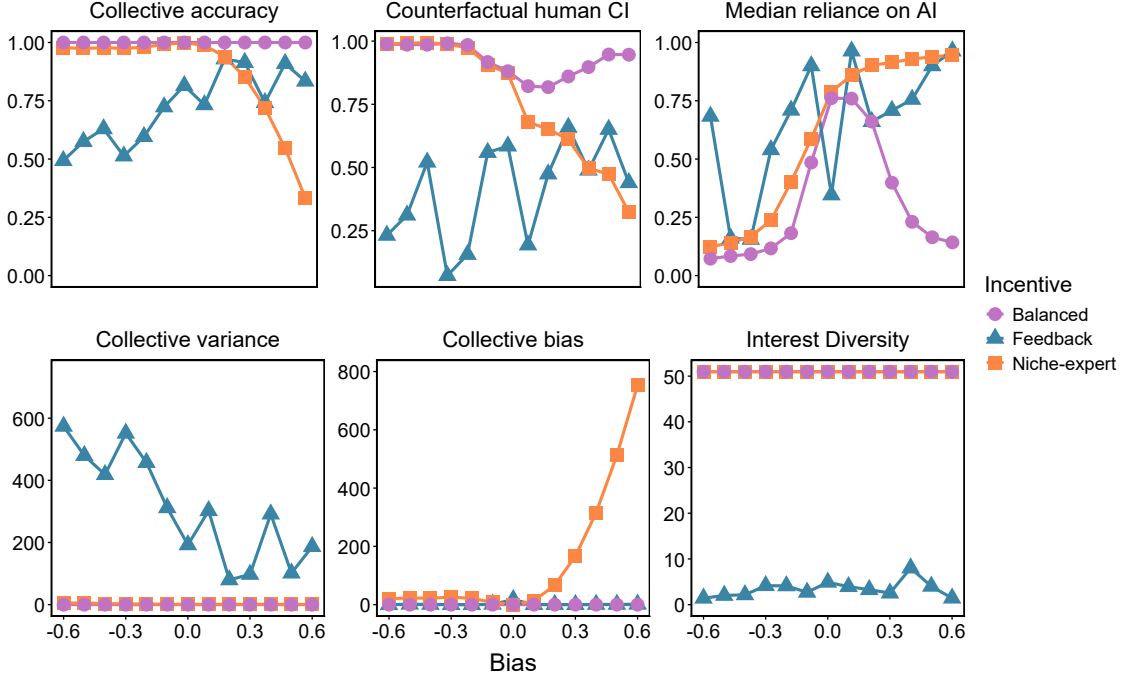


Figure S6: **Comparison of evaluation metrics between incentive structures across various levels of AI bias.** Evaluation metrics under feedback, niche-expert and balanced structures across varying levels of AI bias. Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. All the evaluation metrics are averaged over generations 190,000–200,000. The balanced structure maintains near-perfect collective accuracy across bias levels and higher counterfactual human CI than the other incentive structures. These patterns are driven by well-calibrated reliance on AI, which increases as AI becomes more accurate, and by well-preserved interest diversity, which together lead to low collective variance and collective bias. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, chatbot AI is used.

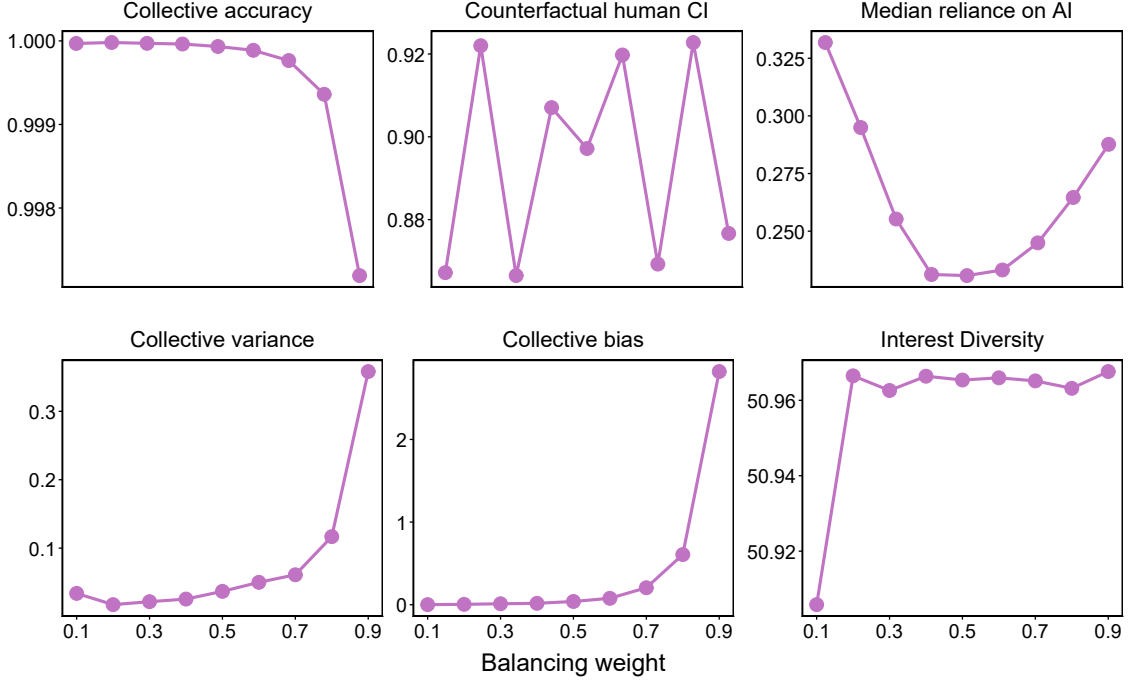


Figure S7: **Balanced structure is robust across varying balancing weights.** Evaluation metrics across varying levels of the balancing weight (0.1 – 0.9). The balancing weight represents the weight assigned to the niche-expert structure when constructing the balanced structure by combining the niche-expert and feedback structures. A balancing weight of 0.5 represents equal weighting of the niche-expert and feedback structures, whereas 0 corresponds to the feedback structure and 1 corresponds to the niche-expert structure. Although the evaluation metrics vary as the balancing weight changes from 0.1 to 0.9, the absolute magnitude of these deviations remains limited. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$, $\tau = 0.3$, chatbot AI is used.

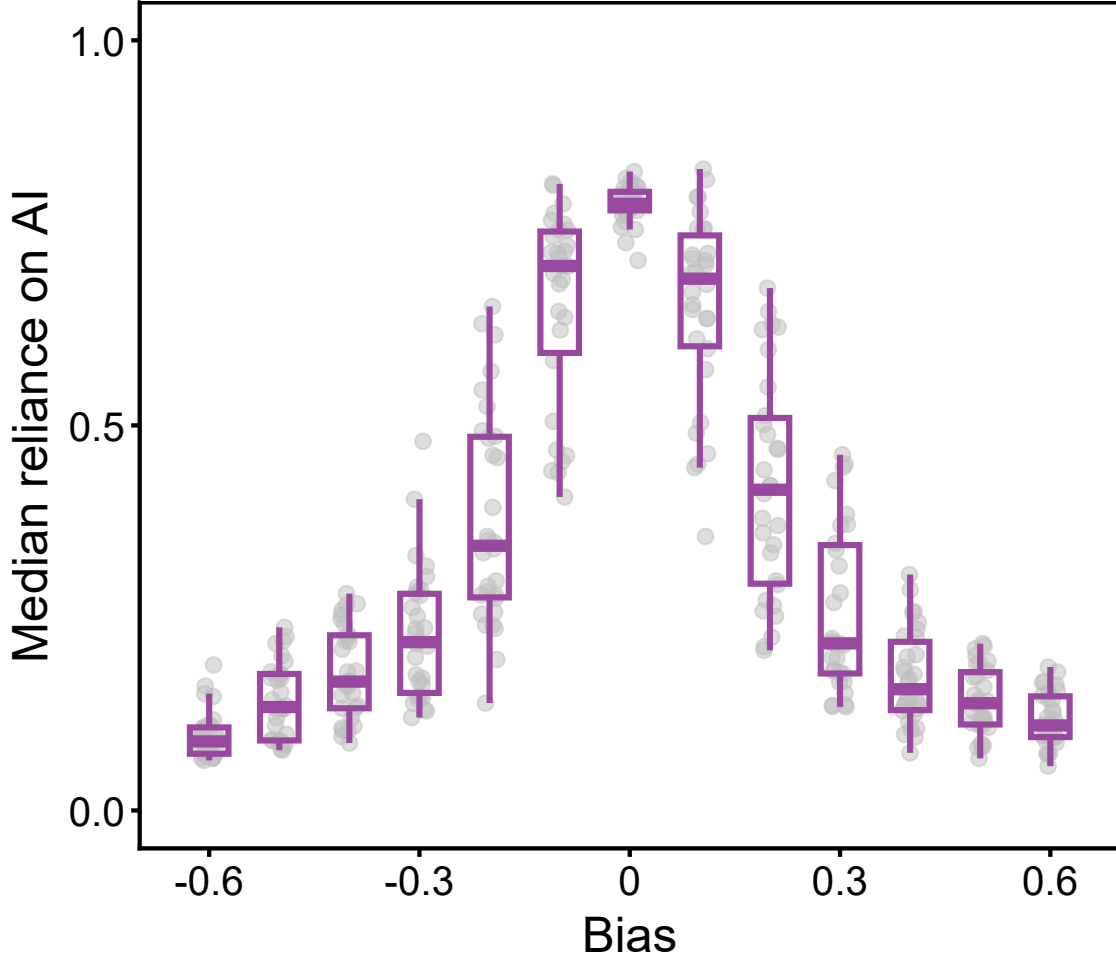


Figure S8: **Balanced structure shows well-calibrated reliance on AI.** Median reliance on AI across varying levels of AI bias. Thirty randomly initialized simulations were performed for each bias level. Each dot represents a single simulation, and the box plots show the median and interquartile range (IQR) for each bias level. The balanced structure shows well-calibrated reliance on AI, with reliance increasing as AI bias decreases. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, chatbot AI is used.

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